

Expected Returns and Dividend Growth Rates Implied by Derivative Markets

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The dividend-price ratio is a noisy proxy for expected returns when expected dividend growth is time-varying. This paper uses a new and forward-looking measure of dividend growth extracted from S&P 500 futures and options to correct the dividend-price ratio for changes in expected dividend growth. Over January 1994 through June 2011, dividend growth implied by derivative markets reliably forecasts future dividend growth, and the corrected dividend-price ratio predicts S&P 500 returns substantially better than the standard dividend-price ratio, in-sample and out-of-sample. Time-varying expected dividend growth is important to explain price movements, especially because it is highly correlated with expected returns. (*JEL* G12, G13, G17)

The predictability of market returns is of great interest to investors and has important implications for asset pricing. At first, any indication of market return predictability was taken as a strike against market efficiency. Since then, with accumulating evidence that price multiples, such as the dividend-price ratio, predict market returns, time-varying expected returns have become an integral part of equilibrium models with efficient markets (Campbell and Cochrane 1999; Bansal and Yaron 2004). Yet return predictability remains debatable. The forecasting relation between returns and the dividend-price ratio is subject to statistical biases (Stambaugh 1999). It is unstable and has poor out-of-sample forecasting power (Goyal and Welch 2008).

This paper argues that the inconsistent evidence in predicting returns is largely due to the time-varying nature of expected dividend growth rates. I introduce a novel measure of expected dividend growth rates that is extracted from index options and futures, and show that the dividend-price ratio corrected

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for variation in expected dividend growth rates predicts market returns both in-sample and out-of-sample.

According to the textbook treatment, the dividend-price ratio is related to expected returns and is thus a good candidate for predicting future returns. It also incorporates information about expected dividend growth rates, however, and if those rates vary over time, the dividend-price ratio is only a noisy proxy for expected returns (Campbell and Shiller 1988; Fama and French 1988). Since the dividend-price ratio increases with expected returns and declines with expected dividend growth rates, the problem is particularly pronounced when expected returns and expected dividend growth are positively correlated (Menzly et al. 2004; Lettau and Ludvigson 2005).

The studies that explicitly account for variation in expected dividends when predicting returns rely on historical dividend data (Kothari and Shanken 1992; Cochrane 2008a; Binsbergen and Koijen 2010; Lacerda and Santa-Clara 2010; Rytchkov 2012). Instead, I extract a proxy for investors' expected dividend growth rates from index options and futures. Prices of these derivatives depend on future dividend payments, and thus provide forward-looking information about expected cash flows. Because index derivatives are highly liquid, new information about future cash flows is rapidly incorporated into their prices.

I motivate my empirical analysis with a simple present value model as in Campbell and Shiller (1988) and Binsbergen and Koijen (2010). Assuming a first-order autoregressive process for expected returns and expected dividend growth rates, I show that the predictable component of returns can be captured through a bivariate return regression on the standard dividend-price ratio and an estimate of expected dividend growth rates, or through a univariate return regression on the corrected dividend-price ratio. The corrected dividend-price ratio is the standard dividend-price ratio adjusted for variation in expected dividend growth rates.

My proxy for expected dividend growth rates, the implied dividend growth rate, is extracted from index derivatives using no-arbitrage relations. Because any mispricing in options is easiest to arbitrage away by trading futures, I combine the cost-of-carry formula for futures and the put-call parity condition for options, and extract dividends simultaneously from options and futures.

I apply the empirical analysis to the S&P 500 index. To balance between dividend seasonalities and the liquidity of derivatives, and to avoid contaminating estimated growth rates by the dividend risk premium, I rely on six-months-to-maturity derivatives. The main analysis runs from January 1994 through June 2011.¹ In this sample, the standard dividend-price ratio is a rather poor predictor of both future returns and dividend growth rates at horizons ranging from one month to one year. Implied dividend growth rates,

¹ Note that, according to the tests of Bai and Perron (1998), this period is not affected by structural breaks in the mean of the dividend-price ratio, which have been shown to affect the forecasting relation of returns and the dividend-price ratio over longer periods (Lettau and Van Nieuwerburgh 2008).

by contrast, are reliable predictors of future dividend growth rates, and the corrected dividend-price ratio is a statistically significant predictor for returns at all horizons.

In monthly return predictive regressions, the corrected dividend-price ratio has an in-sample adjusted R^2 of 4.93% and an out-of-sample R_{OS}^2 of 5.94%, compared with 0.44% and 0.04%, respectively, for the standard dividend-price ratio. Because the corrected dividend-price ratio is less persistent than the standard dividend-price ratio, it has the additional advantage that it is not subject to the Stambaugh (1999) small-sample bias.

Investigating further the implications of the present value model, I find that, in line with the standard view, variation in expected returns is more important for price movements than variation in expected dividend growth rates (Binsbergen and Koijen 2010; Cochrane 2011). Expected dividend growth rates, however, are highly correlated with expected returns, and a large portion of the dividend-price ratio variance can be assigned to the covariance between expected returns and expected dividend growth rates. This covariation makes it hard for the standard dividend-price ratio to predict returns and dividend growth rates (Menzly et al. 2004; Lettau and Ludvigson 2005). It also explains why the corrected dividend-price ratio is more volatile and predicts returns better than the standard dividend-price ratio.

The results are stable across subsamples and are robust to inclusion of other return predictors. They also are not driven by seasonality of dividend payments, or by the risk premium that has been documented to influence dividend prices with longer maturities (Binsbergen et al. 2012 and Binsbergen et al. 2013). The same results cannot be obtained using historical dividend growth rates or information on announced dividends.²

1. Theoretical Considerations

To provide an analytical framework for the empirical analysis, I first derive a simple log-linear present value model. Then I discuss estimation of the dividend growth rates implied by derivative markets.

1.1 Present value model

The model combines the Campbell and Shiller (1988) present value identity with time-series processes for expected returns and expected dividend growth rates. Rytchkov (2012) and Binsbergen and Koijen (2010) use a similar approach. Details are provided in Appendix A.

Define log return r_{t+1} , log dividend growth rate Δd_{t+1} , and log dividend-price ratio dp_t as

$$r_{t+1} = \log \left[\frac{P_{t+1} + D_{t+1}}{P_t} \right], \quad \Delta d_{t+1} = \log \left[\frac{D_{t+1}}{D_t} \right], \quad dp_t = \log \left[\frac{D_t}{P_t} \right]. \quad (1)$$

² In the Online Appendix, I report further robustness checks.

Use first-order Taylor approximation around the long-run average of the dividend-price ratio $\overline{dp}$, and express return as

$$r_{t+1} \simeq \kappa + dp_t + \Delta d_{t+1} - \rho dp_{t+1}, \quad (2)$$

where $\kappa = \log[1 + \exp(-\overline{dp})] + \rho \overline{dp}$ and $\rho = \frac{\exp(-\overline{dp})}{1 + \exp(-\overline{dp})}$. Iterate (2) forward to obtain the Campbell and Shiller (1988) present value identity

$$dp_t \simeq -\frac{\kappa}{1-\rho} + E_t \sum_{j=0}^{\infty} \rho^j (r_{t+1+j}) - E_t \sum_{j=0}^{\infty} \rho^j (\Delta d_{t+1+j}). \quad (3)$$

Let $\mu_t = E_t(r_{t+1})$ be the conditional expected return, and let $g_t = E_t(\Delta d_{t+1})$ be the conditional expected dividend growth rate. Assume that both μ_t and g_t follow $AR(1)$ processes:³

$$\mu_{t+1} = \delta_0 + \delta_1 \mu_t + \varepsilon_{t+1}^{\mu}, \quad (4)$$

$$g_{t+1} = \gamma_0 + \gamma_1 g_t + \varepsilon_{t+1}^g, \quad (5)$$

$$\Delta d_{t+1} = g_t + \varepsilon_{t+1}^d. \quad (6)$$

All errors ε_{t+1}^{μ} , ε_{t+1}^g , and ε_{t+1}^d are zero mean. Combine the present value identity in (3) with the $AR(1)$ assumptions to find the dividend-price ratio

$$dp_t \simeq \varphi + \left(\frac{1}{1-\rho\delta_1} \right) \mu_t - \left(\frac{1}{1-\rho\gamma_1} \right) g_t, \quad (7)$$

where φ is a constant related to $\kappa, \rho, \delta_0, \delta_1, \gamma_0$, and γ_1 .

According to Equation (7), the dividend-price ratio is a function of expected returns and expected dividend growth rates. By the same logic, expected returns are also a function of both the dividend-price ratio and expected dividend growth rates. To derive a return forecasting equation, combine (2), (6), and (7):

$$r_{t+1} \simeq \psi + (1-\rho\delta_1)dp_t + \left(\frac{1-\rho\delta_1}{1-\rho\gamma_1} \right) g_t + v_{t+1}^r, \quad (8)$$

where $v_{t+1}^r = \varepsilon_{t+1}^d - \rho \left(\frac{\varepsilon_{t+1}^{\mu}}{1-\rho\delta_1} - \frac{\varepsilon_{t+1}^g}{1-\rho\gamma_1} \right)$, and ψ is related to $\kappa, \rho, \delta_0, \delta_1, \gamma_0$, and γ_1 .

Alternatively, the forecasting equation can be written as

$$r_{t+1} \simeq \psi + (1-\rho\delta_1)dp_t^{Corr} + v_{t+1}^r, \quad (9)$$

where $dp_t^{Corr} = dp_t + g_t \left(\frac{1}{1-\rho\gamma_1} \right)$ is the corrected dividend-price ratio.

³ The $AR(1)$ structure for expected returns and expected dividend growth rates is considered in several recent contributions (Binsbergen and Koijen 2010; Rytchkov 2012; Piatti and Trojani 2012). Assuming different time-series processes would preserve the main intuition (see Cochrane 2008b for a discussion).

Thus, according to Equations (8) and (9), returns can be predicted either through a bivariate regression on the dividend-price ratio and an estimate for expected dividend growth rates, or through a univariate return regression on the corrected dividend-price ratio. The corrected dividend-price ratio is the dividend-price ratio adjusted for variation in the expected dividend growth rates. The adjustment depends on the linearization parameter ρ and the persistence of expected dividend growth rates γ_1 . The more persistent the expected dividend growth rates, the more important their role in the corrected dividend-price ratio.

Lacerda and Santa-Clara (2010) derive an alternative version of the corrected dividend-price ratio. In their model, only expected returns follow $AR(1)$ process. Consequently, their corrected dividend-price ratio does not depend on persistence of the expected dividend growth rates $dp_t^{LSC} = dp_t + g_t \left(\frac{1}{1-\rho_t} \right)$. Section 4.2 compares results using this alternative correction.

1.2 Estimating implied dividend growth rates

Prices of options and futures depend on, among other factors, future dividend payments, and are related through the laws of no-arbitrage. Assuming no arbitrage opportunities, an estimate for expected dividend growth rates can therefore be extracted from the market prices of derivatives.

Following Ait-Sahalia and Lo (1998), I assume that the index pays a continuously compounded dividend yield. I start with the case of a known dividend yield, and then move on to a stochastic dividend yield.

The no-arbitrage futures price is given by the cost-of-carry formula

$$F_t(\tau) = S_t \exp[(r_t(\tau) - \lambda_t(\tau))\tau], \quad (10)$$

where F_t is the future's price, S_t is the price of the underlying, $r_t(\tau)$ is the annualized continuously compounded interest rate from t to $t+\tau$, and $\lambda_t(\tau)$ is the annualized continuously compounded dividend yield between t and $t+\tau$. Similarly, by no-arbitrage, the difference between a European call and a put option written on the index is given by put-call parity

$$C_t(K, \tau) - P_t(K, \tau) = S_t \exp[-\lambda_t(\tau)\tau] - K \exp[-r_t(\tau)\tau], \quad (11)$$

where $C_t(K, \tau)$ and $P_t(K, \tau)$ are the prices of a European call and a put option with the same maturity τ and the same strike price K .

Both no-arbitrage conditions relate prices of derivatives to the future dividend yield and the risk-free rate. With a proxy for the risk-free rate, dividend yield can therefore be extracted from either options or futures. The interest rate implied by derivatives, however, may differ from the observable proxies for the interest rate (Naranjo 2009). Also, given that the basket of stocks in the index is difficult to trade, any mispricing of options is easiest to arbitrage away by trading futures (Kamara and Miller 1995). Therefore, I propose to combine

both no-arbitrage relations, and extract the dividend yield from index options and index futures simultaneously.⁴

I first solve for the interest rate implied by derivative markets by expressing Equation (10) in terms of the price of the underlying and inserting it into (11):

$$r_t(\tau) = \frac{1}{\tau} \log \left[\frac{F_t(\tau) - K}{C_t(K, \tau) - P_t(K, \tau)} \right]. \quad (12)$$

Then I plug (12) back into (11) to obtain a formula for the dividend yield implied by derivative markets:

$$\lambda_t(\tau) = -\frac{1}{\tau} \log \left[\left(\frac{C_t(K, \tau) - P_t(K, \tau)}{S_t} \right) + \frac{K}{S_t} \left(\frac{C_t(K, \tau) - P_t(K, \tau)}{F_t(\tau) - K} \right) \right]. \quad (13)$$

Following Equation (13), all I need to estimate the implied dividend yield is a European call option and a European put option with the same strike and the same maturity, the futures price with the same expiration date as the options, and the price of the underlying asset.

Equation (13), however, is derived under the assumption of a known dividend yield. When the dividends are stochastic, as assumed in the present value model, the dividend yield not only reflects expectations about the future dividend yield, but may also include a dividend risk premium (Binsbergen et al. 2012, 2013). To avoid such contamination, as discussed below, I choose options and futures with six months to maturity, where a substantial portion of dividends is preannounced, and hence the effect of the risk premium is mitigated.

Because the implied dividend yield $\lambda_t(\tau)$ is continuously compounded, I transform it into a raw (effective) implied dividend yield, $IDY_t = \exp(\hat{\lambda}_t) - 1$. The proxy for the expected dividend growth (implied dividend growth) is then the difference between the log-implied dividend yield and the log dividend-price ratio $idg_t = \log(IDY_t) - dp_t$.

2. Data Description

In this section, I first describe the data sources, define the main variables, and detail the empirical approach to estimating implied dividend growth rates. Then I discuss the main characteristics of the estimated implied dividend growth rates and the corrected dividend-price ratio.

2.1 Data sources

I use the S&P 500 index as a proxy for the aggregate market. The daily S&P 500 price index (without dividends) and the total return index (with dividends reinvested) are from Datastream. The daily S&P 500 futures data are from the

⁴ In the Online Appendix, I consider alternative ways to estimate implied dividend yields and find that results are indeed the strongest when implied dividend yields are estimated simultaneously from futures and options.

Chicago Mercantile Exchange, and the end-of-day S&P 500 options data are from Market Data Express.

Data for control variables come from several sources. The default spread and the term spread are from the Federal Reserve Bank of St. Louis. The S&P 500 earnings-price ratio and the six-month LIBOR are from Datastream. Implied skewness comes from the Chicago Board Options Exchange (CBOE). End-of-month variance risk premium estimates are from Hao Zhou's Web page, and the quarterly consumption-to-wealth ratio is from Sydney C. Ludvigson's Web page. SPX OpenClose data on opening and closing buy and sell volume for different investor groups are from Market Data Express. Dividends and prices over longer time periods are from Amit Goyal's Web page. Indicated dividend series for the S&P 500 index are from Standard & Poor's.

European options on the S&P 500 index have been traded since April 1986. The Market Data Express options data, however, go back only to January 1990. Furthermore, while futures are settled at the opening value of the index, the most liquid S&P 500 options initially expired at the closing value of the index. As liquid options and futures with matching expiration times are needed to estimate implied dividend growth rates, I confine the main analysis to the January 1994–June 2011 period.⁵

I eliminate all nonstandard S&P 500 options (quarterlies, weeklies, and mini options) as well as options that violate the basic arbitrage relations. All variables are sampled at the end of the month.

2.2 Variable definitions and empirical estimation

Monthly returns. Monthly returns are defined as

$$r_t^M = \log \left[\frac{P_t + D_t}{P_{t-1}} \right], \quad (14)$$

where P_t is price at the end of month t , and D_t is the sum of daily dividends in month t . Daily dividends are calculated as the difference between the return on the total return index and the return on the price index multiplied by the lagged price index.

Standard dividend-price ratio. The dividend-price ratio is a log ratio of a twelve-month trailing sum of dividends D_t^{12} and the current price

$$dp_t = \log \left[\frac{D_t^{12}}{P_t} \right]. \quad (15)$$

Following Chen (2009) and Koijen and Van Nieuwerburgh (2011), I do not reinvest dividends during the year to make sure that the market return does not contaminate the dividend series.

⁵ In the Online Appendix, I consider extending the time period back to 1990.

Dividend growth rates. As in Ang and Bekaert (2007), I calculate monthly dividend growth rates based on a twelve-month trailing sum of dividends

$$\Delta d_t^M = \log \left[\frac{D_t^{12}}{D_{t-1}^{12}} \right]. \quad (16)$$

This approach to calculating dividend growth rates diminishes the effect of seasonality in dividend payments, but introduces overlapping for any frequency higher than annual.

Implied dividend growth rates. As defined above, implied dividend growth rates are the difference between the log-implied dividend yield and the log dividend-price ratio

$$idg_t = \log(IDY_t) - dp_t, \quad (17)$$

where $IDY_t = [\exp(\hat{\lambda}_t) - 1]$, and the continuously compounded implied dividend yield $\lambda_t(\tau)$ is estimated according to Equation (13).

I estimate implied dividend yields using daily settlement prices for futures, the midpoint between the last bid and the last ask price for options, and closing values for the S&P 500 price index.⁶ To minimize effects of microstructure noise, I aggregate information from a wide set of options and futures. At the end of each month, I use ten days of data (backward-looking) and construct put-call pairs from all options with positive volume or open interest greater than 200 contracts.⁷ Then I combine these option pairs with futures of matching maturity. Because options expire monthly and futures expire quarterly, I use only derivatives with quarterly expirations (first three closest quarterly expirations). To eliminate some extreme values, I discard observations where $\frac{F_t(\tau) - K}{C_t(K, \tau) - P_t(K, \tau)}$ is lower than 0.5 or higher than 1.5, and where $C_t(K, \tau) - P_t(K, \tau)$ is zero.⁸

This matching procedure results in 145,874 estimated dividend yields (227 on average per maturity per month). The number of estimates increases over time and declines with the maturity. The overall mean of the estimates is approximately the same across maturities, 0.017 on average. The dispersion of the estimates, however, declines with the maturity. The interquartile range of the estimates based on the first quarterly expiration is 0.057, compared with 0.017 and 0.013 for the second and the third quarterly expirations, respectively. I aggregate these estimates at the end of each month by taking the median across all implied dividend yields with the same maturity (see also Binsbergen et al. 2012). To obtain monthly implied dividend yields with constant maturities, I interpolate linearly between yields with closest maturities.

⁶ In the Online Appendix, I show that the results are not driven by nonsynchronicity issues in the end-of-day data.

⁷ In untabulated results, I show that there is no strike price effect; i.e., the implied dividend yield does not depend on the moneyness level.

⁸ This filter eliminates under 2% of the observations.

Choice of the most appropriate maturity of the implied dividend yield depends on several considerations. To alleviate within-year seasonality of dividends, the analysis should ideally be based on derivatives with one year to expiration. As derivatives are most liquid at the short end, however, implied dividend yields estimated from near-to-maturity options and futures provide more reliable estimates. Also, the shorter the maturity of the implied dividend yield, the less the effects of the risk premium. Choice of the maturity of the implied dividend yield therefore must balance three aspects: the dividend risk premium, the within-year seasonality of dividend payments, and the liquidity of derivatives.

First, Binsbergen et al. (2012) document that the dividend risk premium plays an important role for dividend claims of maturities between one and two years.⁹ They also find, however, that a trading strategy that involves holding two years of dividends and a steepener strategy that sells off the first six months of dividends yield similar risk premiums. This suggests that the risk premium for short-maturity dividend claims is low. Such a result is intuitive because at short maturities a substantial portion of dividends is preannounced and hence riskless.¹⁰ To test for the presence of the dividend risk premium in the period analyzed in this paper, I consider a trading strategy that goes long a six-month dividend claim, collects dividends through the month, sells the claim after a month, and then again goes long a new six-month dividend claim. The log return on such a trading strategy is 0.96% annually, lower than the risk-free rate.¹¹ Thus, there is no evidence of risk premium in the short-horizon dividend claims in the analyzed period.

Second, in Table 1, I examine seasonalities in realized dividends. Panel A collects the summary statistics for annualized dividends summed over one month, six months, and twelve months. Panel B reports the summary statistics for dividend growth rates defined as six-month over six-month dividends $\Delta d_t(6m/6m) = \log(D_t^6/D_{t-6}^6)$, six-month over twelve-month dividends $\Delta d_t(6m/12m) = \log(2 \times D_t^6/D_{t-6}^{12})$, and twelve-month over twelve-month dividends $\Delta d_t(12m/12m) = \log(D_t^{12}/D_{t-12}^{12})$.

According to Panel A, annualized monthly dividends are 60% more volatile than annual dividends and are thus highly seasonal. Annualized dividends summed over six months by contrast are only marginally more volatile than annual dividends. Any remaining seasonalities in the six-month dividends, as reported in Panel B, are further reduced by including annual dividends in

⁹ A similar conclusion is drawn by Binsbergen et al. (2013), who analyze a growing market for dividend futures. Dividend futures allow the trading of dividends directly, and are thus suitable for the study of implied dividends. The disadvantage is the short time period. They were introduced around 2000 and have been exchange traded only since 2008.

¹⁰ In Section 6, I show that a sizable part of the dividend growth can be predicted using information on announced dividends.

¹¹ One dollar invested in this trading strategy in January 1994 grows to \$1.18 in June 2011. By comparison, one dollar invested in the risk-free rate grows to \$1.78.

Table 1
Seasonality of realized dividends

Panel A: Dividend levels

	<i>Div-1M</i>	<i>Div-6M</i>	<i>Div-12M</i>
Mean	19.2090	19.2051	19.2169
Min	6.8514	12.8991	13.1692
Max	41.7732	29.7201	29.0365
Stdev	7.3883	4.7054	4.6176

Panel B: Dividend growth rates

	$\Delta d_t(6m/6m)$	$\Delta d_t(6m/12m)$	$\Delta d_t(12m/12m)$
Mean	0.0207	0.0297	0.0386
Min	-0.2319	-0.2347	-0.2224
Max	0.1700	0.1611	0.1577
Stdev	0.0639	0.0704	0.0824
AR(1)	0.7415	0.9117	0.9776
AR(6)	-0.0608	0.5405	0.7868
AR(12)	0.4300	0.3040	0.3920

Panel A reports summary statistics for annualized S&P 500 dividends aggregated over one month *Div-1M*, six months *Div-6M*, or twelve months *Div-12M*. Panel B reports summary statistics for dividend growth rates defined as six-month over six-month dividends $\Delta d_t(6m/6m) = \log(D_t^6/D_{t-6}^6)$, six-month over twelve-month dividends $\Delta d_t(6m/12m) = \log(2 \times D_t^6/D_{t-6}^{12})$, and twelve-month over twelve-month dividends $\Delta d_t(12m/12m) = \log(D_t^{12}/D_{t-12}^{12})$. The period is from January 1994 through June 2011, and the frequency is monthly.

the denominator of the growth rates. In particular, while the autocorrelation coefficients of the $\Delta d_t(6m/6m)$ have alternating signs, the autocorrelation coefficients of the $\Delta d_t(6m/12m)$ and the $\Delta d_t(12m/12m)$ exhibit similar decaying patterns. Hence, seasonalities in growth rates involving six-month dividends are largely diminished when the denominator is based on a twelve-month trailing sum of dividends.

Third, Table 2 reports daily open interest and volume for S&P 500 options and futures. The short-maturity derivatives are the most liquid. The tilt toward short maturities is especially pronounced for futures. The markets for S&P 500 options and futures, however, are among the most liquid derivative markets, and there is a considerable amount of activity in the longer-maturity derivatives as well. For example, the average daily open interest of futures with the second-(third-)closest quarterly expiration amounts to \$5.93 billion (\$839 million), respectively. Thus, there is a substantial amount of trading in S&P 500 derivatives with maturities up to six months.

Given this analysis, I choose to conduct the main analysis using options and futures with six months to expiration. At this maturity the dividend risk premium is absent, derivatives are liquid enough to provide reliable estimates, and seasonality is not a concern because, according to the definition of the implied dividend growth rates, dividend yield extracted from derivatives is standardized by the annual dividend-price ratio.¹²

¹² In the Online Appendix, I show that any remaining seasonalities do not affect my main predictability results.

Table 2
Liquidity of S&P 500 derivatives

Quarterly expirations (maturity T in months)	First-closest ($T \leq 3$)	Second-closest ($3 < T \leq 6$)	Third-closest ($6 < T \leq 9$)	All others ($9 < T$)
Panel A: Options				
Volume in number of contracts	84,561	14,418	7,338	6,075
Open interest in number of contracts	1,423,791	595,876	363,913	380,895
Volume in notional value	9,785,820,724	1,659,107,660	846,344,457	712,968,437
Open interest in notional value	162,676,482,320	67,976,582,059	42,304,187,941	44,640,651,681
Panel B: Futures				
Volume in number of contracts	54,497	2,390	142	35
Open interest in number of contracts	414,146	21,353	3,132	814
Volume in notional value	14,037,995,757	662,610,862	39,141,370	8,906,572
Open interest in notional value	116,944,132,142	5,930,752,412	839,421,108	231,160,321

This table reports average daily open interest and volume by maturity for S&P 500 options and futures. Only options and futures with matching quarterly expirations are included. Open interest and volume are expressed either in number of contracts or in notional value (in \$). Notional value for open interest (volume) is calculated as open interest (volume) times the spot price of the index times the multiplier. Options multiplier is 100, and futures multiplier is 250. The period is from January 1994 through June 2011.

Corrected dividend-price ratio. The corrected dividend-price ratio is

$$dp_t^{Corr}(h) = dp_t + \left(\frac{1}{1 - \widehat{\rho\gamma_1}(h)} \right) idg_t = dp_t + \widehat{AF}(h) \times idg_t, \quad (18)$$

where h is horizon in months. $\widehat{AF}(h)$ is the adjustment factor that depends on the estimated linearization constant $\widehat{\rho}$ and the persistence of the implied dividend growth rates $\widehat{\gamma_1}(h)$. Persistence of the expected dividend growth rates is measured over the same horizon as returns.

2.3 Main characteristics of the implied dividend growth rates and the corrected dividend-price ratio

Figure 1 depicts the implied dividend growth rates (Panel A) and the standard dividend-price ratio along with the annual corrected dividend-price ratio (Panel B). Summary statistics for the main variables are collected in Table 3.

The implied dividend growth rates are 7.31% on average. This is higher than the average for the realized dividend growth rates of 3.77%, but the difference is not significant. The implied dividend growth rates also nicely reflect general market conditions. They are positive during the market booms of 1994–1997 and 2002–2007, when investors were optimistic about future growth opportunities, and generally negative during stock market busts such as the Asian-Russian-LTCM crisis (1998), the dot.com bubble burst (2001), and the recent financial crisis (2008–2009), when investors were pessimistic about growth opportunities.

A somewhat disconcerting feature is the rather high volatility of the implied dividend growth rates, especially around the turn of the century. These

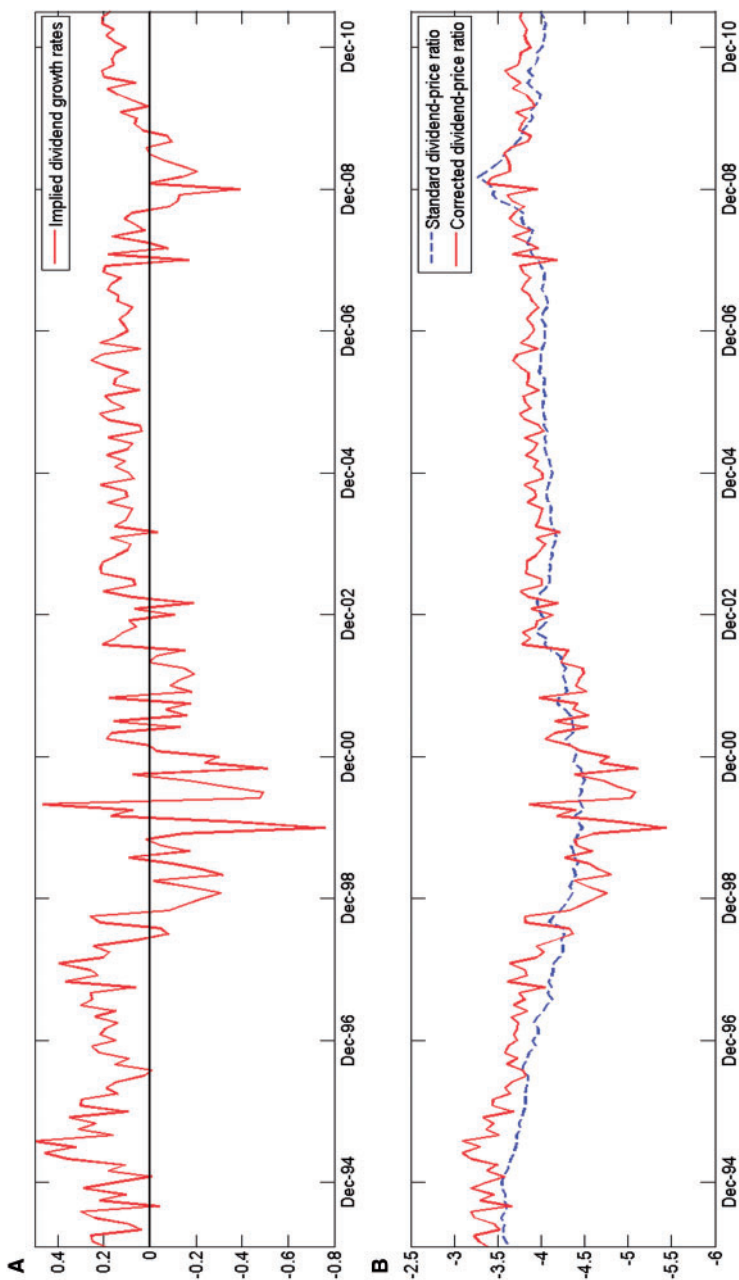


Figure 1
Implied dividend growth rates and dividend ratios

This figure plots the implied dividend growth rates (Panel A), and the standard dividend-price ratio along with the annual corrected dividend-price ratio (Panel B) for the S&P 500 index. The period is from January 1994 through June 2011.

Table 3
Summary statistics

	r_t^M	Δd_t^M	idg_t	dp_t	$dp_t^{Corr(1)}$	$dp_t^{Corr(3)}$	$dp_t^{Corr(6)}$	$dp_t^{Corr(12)}$
Panel A: Summary statistics								
Mean	0.0784	0.0377	0.0731	-4.0150	-3.4763	-3.7941	-3.8807	-3.9223
Stdev	0.1570	0.0416	0.1810	0.2580	1.4349	0.6731	0.4798	0.3947
AR (1)	0.1107	0.1033	0.5334	0.9780	0.5910	0.6793	0.7584	0.8216
ADF Test	-10.0492	-8.0882	-5.5914	-2.1315	-5.1022	-4.3915	-3.7697	-3.2681
Panel B: Unconditional correlations								
r_t^M	1.0000	-0.0323	0.0705	-0.0773	0.0516	0.0277	0.0073	-0.0095
Δd_t^M		1.0000	0.1704	-0.0866	0.1429	0.1053	0.0716	0.0425
idg_t			1.0000	0.3088	0.9853	0.9312	0.8593	0.7833
dp_t				1.0000	0.4669	0.6343	0.7518	0.8331
$dp_t^{Corr(1)}$					1.0000	0.9798	0.9341	0.8781
$dp_t^{Corr(3)}$						1.0000	0.9866	0.9560
$dp_t^{Corr(6)}$							1.0000	0.9910
$dp_t^{Corr(12)}$								1.0000

This table reports summary statistics (Panel A) and unconditional correlations (Panel B) for S&P 500 monthly returns r_t^M , monthly dividend growth rates Δd_t^M , implied dividend growth rates idg_t , standard dividend-price ratio dp_t , and four different versions of the corrected dividend-price ratio corresponding to different return horizons h (in months), $dp_t^{Corr(h)} = dp_t + \widehat{AF(h)} \times idg_t$, where $\widehat{AF(h)}$ for $h = 1, 3, 6$, and 12 are 7.37, 3.02, 1.84, and 1.27, respectively. All variables are expressed in logs. Sample means and standard deviations (Stdev) for monthly returns and monthly realized dividend growth rates are multiplied by 12 ($\sqrt{12}$). An augmented Dickey-Fuller test (ADF test) tests the null hypothesis of a unit root. The critical values corresponding to p -values of 0.01, 0.05, and 0.10 are -3.99, -3.42, and -3.14, respectively. The period is from January 1994 through June 2011, and the frequency is monthly.

fluctuations could be driven partially by data problems.¹³ Because implied dividend yields are estimated from no-arbitrage relations that involve highly leveraged positions, any errors would be magnified in the implied dividend growth rates (Boguth et al. 2012). Importantly, however, as documented in Section 3.1, implied dividend growth rates reliably predict future dividend growth rates.

Any measurement errors in the implied dividend growth rates, nevertheless, would bias the persistence estimates downward. Assuming that the measurement errors are random, simulations in Appendix B show that there can be a substantial downward bias in the monthly persistence of the implied dividend growth rates. Because such bias declines with the length of the horizon, I infer higher-frequency persistence from its annual values. This approach also circumvents the problem of overlapping observations. I exploit the fact that the data are available monthly and take the average of the $AR(1)$ across 12 successive annual samples. I find an annual $AR(1)$ of 0.22, which implies a monthly $AR(1)$ of 0.88.¹⁴

¹³ Golez and Jackwerth (2012) note an unusually high number of missing observations of S&P 500 options in 1998–2002 in both Market Data Express and OptionMetrics data.

¹⁴ Thus, the monthly persistence of the implied dividend growth rates is much higher than suggested by the simple $AR(1)$ of 0.53 reported in Table 3, which does not take into account bias due to measurement errors (and overlapping observations).

Using these persistence estimates, the estimated adjustment factors $\widehat{AF(h)}$ in the corrected dividend-price ratio (Equation 18) for horizons $h=1, 3, 6$, and 12 months are 7.37, 3.02, 1.84, and 1.27, respectively. Thus, the influence of the implied dividend growth rates on the corrected dividend-price ratio weakens with the length of the return horizon.¹⁵

All versions of the corrected dividend-price ratio are strongly correlated with the standard dividend-price ratio, with a pairwise correlation coefficient between 0.47 and 0.83. Their patterns, however, differ from the standard dividend-price ratio in three important aspects.

First, the corrected dividend-price ratios are on average higher than the standard dividend-price ratio in boom periods and lower in bust periods. This means that the standard dividend-price ratio tends to predict returns that are too low to match the market's optimism about growth opportunities during the boom periods. Similarly, the standard dividend-price ratio tends to predict returns that are too high during the crisis periods.

Second, the corrected dividend-price ratios are notably more volatile than the standard dividend-price ratio. The volatility is 0.26 for the standard dividend-price ratio compared with 1.43 and 0.39 for the corrected dividend-price ratio corresponding to monthly and annual return horizons, respectively. While measurement errors in the implied dividend growth rates could account for part of this difference, in the present value model, such an increase in volatility implies that expected returns and expected dividend growth rates are highly positively correlated. This positive correlation offsets shocks to expected returns and to expected dividend growth rates, and reduces the volatility of the dividend-price ratio. Adjusting the dividend-price ratio for expected dividend growth rates then restores part of the variation in the dividend-price ratio, which is otherwise offset by this positive correlation (see also Lacerda and Santa-Clara 2010).¹⁶

Finally, the corrected dividend-price ratios are also less persistent than the standard dividend-price ratio. The monthly $AR(1)$ for the standard dividend-price ratio is 0.98, and the null hypothesis of a unit root cannot be rejected, according to an augmented Dickey-Fuller (ADF) test. The $AR(1)$ for the corrected dividend-price ratios is between 0.59 and 0.82, and the null of a unit root in the corrected dividend-price ratios is rejected at least at the 5%

¹⁵ Because $\hat{\rho}$ and $\hat{\gamma}_1$ are estimated using the whole sample, the corrected dividend-price ratio is subject to a look-ahead bias. In Section 4.1 I show, however, that look-ahead bias has only a marginal influence on the results.

¹⁶ To see this formally, expand the variance of the corrected dividend-price ratio as

$$var(dp_t^{Corr}) = var(dp_t) + 2 \left(\frac{1}{1-\rho\gamma_1} \right) \left(\frac{1}{1-\rho\delta_1} \right) cov(\mu_t, g_t) - \left(\frac{1}{1-\rho\gamma_1} \right)^2 var(g_t). \quad (19)$$

Hence, $var(dp_t^{Corr}) > var(dp_t)$ only if expected returns and expected dividend growth rates covary and the covariation is strong enough $\left(2 \left(\frac{1}{1-\rho\delta_1} \right) cov(\mu_t, g_t) > \left(\frac{1}{1-\rho\gamma_1} \right) var(g_t) \right)$.

level. This reduced persistence is important because it mitigates the effect of the small-sample bias (Stambaugh 1999).

3. Results

The present value model implies that all available information is incorporated in short-term (one-horizon) prices and dividends. The model is silent, however, about the appropriate length of the horizon. I consider predicting dividend growth rates and market returns from one month to one year. I report Hodrick (1992) t -statistics and adjusted R^2 s. The t -statistics take into account the overlapping structure of regressions.¹⁷ The R^2 s in the context of overlapping observations, however, need to be interpreted with caution because they tend to increase with the extent of the overlap even in the absence of true predictability (Valkanov 2003; Boudoukh et al. 2008).

3.1 Predicting dividend growth rates

The main dividend growth rate regression is

$$\Delta d_{t+h} = a_0 + a_1 X_t + \varepsilon_{t+h}, \quad (20)$$

where $\Delta d_{t+h} = \sum_{i=1}^h \Delta d_{t+i}^M$ is the dividend growth rate with $h=1, 3, 6$, or 12 months, and X_t is a set of predictor variables that includes implied dividend growth rates, the standard dividend-price ratio, and lagged dividend growth rates. Because of the overlapping structure in the realized dividend growth rates, I use lagged annual dividend growth rates only in predictive regressions for annual dividend growth rates.

Table 4 reports the results. The standard dividend-price ratio is negatively related to future dividend growth rates—as the theory suggests—but the associated t -statistics are relatively low and vary between -1.87 and -1.54 . Accordingly, the associated $adj. R^2$ s are low, ranging from 1.57% for predicting monthly dividend growth rates to 4.40% for predicting annual dividend growth rates. The implied dividend growth rates, by contrast, are positively related to future dividend growth rates, explaining 5.14% of the variation in monthly dividend growth rates and 19.16% of the variation in annual dividend growth rates. All the estimated parameters on the implied dividend growth rates are highly significant, with t -statistics varying between 3.76 and 4.98.

The implied dividend growth rates remain significant in bivariate predictive regressions with the standard dividend-price ratio, which confirms that the implied dividend growth rates provide an independent source of information about future dividend growth rates. Moreover, in bivariate regressions, the

¹⁷ Ang and Bekaert (2007) show that the Hodrick (1992) standard errors, which are based on summing predictors in the past, perform better than other standard errors frequently employed in the literature, such as Newey-West (1987) or Hansen and Hodrick (1980) standard errors.

Table 4
Dividend growth rate regressions

Dividend growth horizon h	1	3	6	12
dp_t	-0.0067 (-1.87)	-0.0192 (-1.73)	-0.0390 (-1.69)	-0.0681 (-1.54)
$adj.R^2$	0.0157	0.0412	0.0492	0.0440
idg_t	0.0157 (3.76)	0.0492 (4.39)	0.1034 (4.98)	0.1957 (4.82)
$adj.R^2$	0.0514	0.1424	0.1809	0.1916
dp_t	-0.0111 (-2.70)	-0.0331 (-2.58)	-0.0681 (-2.55)	-0.1216 (-2.41)
idg_t	0.0206 (4.09)	0.0639 (4.55)	0.1338 (4.92)	0.2496 (4.85)
$adj.R^2$	0.0990	0.2621	0.3268	0.3301
idg_t				0.1674 (4.50)
Δd_t^A				0.2920 (2.50)
$adj.R^2$				0.2645

This table reports regression results for predicting S&P 500 dividend growth rates. Predictor variables include the standard dividend-price ratio dp_t , the implied dividend growth rates idg_t , and the annual realized dividend growth rates Δd_t^A . Dividend growth rate horizon h is in months. All variables are expressed in logs. Hodrick (1992) t -statistics are reported in parentheses. All regressions include a constant term. The period is from January 1994 through June 2011, and the frequency is monthly.

statistical significance of both predictors (the implied dividend growth rates and the standard dividend-price ratio) is higher than in the univariate regressions, and the $adj.R^2$ rises to 9.90% in a regression for monthly dividend growth rates and to 33.01% in a regression for annual dividend growth rates.¹⁸

The implied dividend growth rates also remain highly significant in the bivariate regression with realized dividend growth rates. While the t -statistic associated with the realized dividend growth rates in the bivariate regression is 2.50, it is 4.50 for implied dividend growth rates.

3.2 Predicting market returns

The main return regression is

$$r_{t+h} = b_0 + b_1 X_t + \varepsilon_{t+h}, \tag{21}$$

where $r_{t+h} = \sum_{i=1}^h r_{t+i}^M$ is the market return with $h=1, 3, 6$, or 12 months, and X_t is a set of predictor variables that includes the standard dividend-price ratio, the implied dividend growth rates, the corrected dividend-price ratio, and the

¹⁸ This improvement in predicting dividend growth rates can be accounted for by partial resolution of the omitted variable bias. According to the present value model, a regression of dividend growth rates on the dividend-price ratio is subject to an omitted variable bias because it lacks a control for expected returns. When expected dividend growth rates are highly correlated with expected returns, a proxy for expected dividend growth rates (implied dividend growth rates) can help to diminish the omitted variable bias. That omitted variable bias plays an important role in the dividend dividend growth rate predictive regressions is documented in Binsbergen and Koijen (2010) and Koijen and Van Nieuwerburgh (2011). See also discussion in Ang and Bekaert (2007).

Table 5
Return regressions

Return horizon h	1	3	6	12
dp_t	0.0168 (1.15)	0.0593 (1.41)	0.1408 (1.78)	0.3159 (2.21)
$adj.R^2$	0.0044	0.0292	0.0767	0.1763
dp_t	0.0047 (0.29)	0.0395 (0.85)	0.1164 (1.36)	0.2768 (1.88)
idg_t	0.0558 (2.55)	0.0915 (1.76)	0.1119 (1.27)	0.1826 (1.24)
$adj.R^2$	0.0449	0.0604	0.0949	0.1988
$dp_t^{Corr(h)}$	0.0073 (3.22)	0.0325 (2.61)	0.0835 (2.43)	0.2167 (2.54)
$adj.R^2$	0.0493	0.0645	0.0939	0.1931
dp_t	0.0036 (0.21)	0.0372 (0.78)	0.1128 (1.28)	0.2707 (1.79)
idg_t	0.0587 (2.53)	0.0976 (1.78)	0.1215 (1.29)	0.2009 (1.35)
Δd_t^A	-0.0273 (-0.65)	-0.0575 (-0.46)	-0.0908 (-0.36)	-0.1607 (-0.37)
$adj.R^2$	0.0424	0.0587	0.0934	0.1986

This table reports regression results for predicting S&P 500 returns. Predictor variables include the standard dividend-price ratio dp_t , the implied dividend growth rates idg_t , the annual realized dividend growth rates Δd_t^A , and the corrected dividend-price ratio defined as $dp_t^{Corr(h)} = dp_t + \widehat{AF(h)} \times idg_t$, where $\widehat{AF(h)}$ for $h=1, 3, 6$, and 12 are 7.37, 3.02, 1.84, and 1.27, respectively. Return horizon h is in months. All variables are expressed in logs. Hodrick (1992) t -statistics are reported in parentheses. All regressions include a constant term. The period is from January 1994 through June 2011, and the frequency is monthly.

lagged dividend growth rates. Note that, as discussed above, the corrected dividend-price ratio varies with the return horizon.

Table 5 reports the results. In accordance with the theory, the estimated parameter on the standard dividend-price ratio is positive, but the associated t -statistics are relatively low, especially at the short end. The estimated parameter is significant at the conventional 5% significance level only in the predictive regression for annual returns. The associated $adj.R^2$ s vary from 0.44% for predicting monthly returns to 17.63% for predicting annual returns.

When the implied dividend growth rates are included as an additional regressor, return predictability improves for all the horizons considered. The $adj.R^2$ rises from 0.44% to 4.49% in the regression for monthly returns and from 17.63% to 19.88% in the regressions for annual returns.

Importantly, the corrected dividend-price ratio predicts returns approximately as well as the standard dividend-price ratio and the implied dividend growth rates together. The $adj.R^2$ amounts to 4.93% at the monthly horizon and to 19.31% at the annual horizon. Furthermore, estimated parameters on the corrected dividend-price ratios are always statistically significant, with Hodrick (1992) t -statistics varying between 2.43 and 3.22. In untabulated results, I show that the F -test fails to reject restriction implied by the present value model on the relation between the dividend-price ratio and the implied dividend growth rates, $dp_t^{Corr(h)} = dp_t + \widehat{AF(h)} \times idg_t$, at all the considered horizons.

The corrected dividend-price ratio is also not subject to the Stambaugh (1999) bias. Untabulated simulations show a very low upward bias in the estimated predictive coefficients, 1%–4%, depending on the return horizon. The associated bootstrapped p -values are between 0% and 1%.¹⁹ In comparison, bias in the standard dividend-price ratio can be as high as 100%, and all the bootstrapped p -values on the standard dividend-price ratio are insignificant, between 29% and 48%.

Finally, the same results cannot be obtained using lagged dividend growth rates instead of implied dividend growth rates. Inclusion of realized dividend growth rates in the return predictive regressions with the standard dividend-price ratio and the implied dividend growth rates leaves the $adj.R^2$ largely unchanged. The realized dividend growth rates are statistically insignificant at all horizons.

4. Additional Tests

In this section, I examine the stability of in-sample forecasting relations and consider predicting returns out-of-sample. I also analyze how the corrected dividend-price ratio compares with other return predictors. To avoid statistical problems inherent in the use of overlapping observations (Kirby 1997; Valkanov 2003; Boudoukh et al. 2008), I restrict the analysis henceforth to prediction of nonoverlapping monthly returns.

4.1 Parameter stability and out-of-sample predictability

To analyze stability of in-sample forecasting relations, Figure 2 plots predictive coefficients (Panel A) and t -statistics (Panel B) for predictive regressions of monthly returns on either the standard dividend-price ratio or the corrected dividend-price ratio using 120-month rolling windows.

The forecasting relation between returns and the standard dividend-price ratio is very unstable. The predictive coefficient on the standard dividend-price ratio even became negative during the recent financial crisis. The predictive coefficient on the corrected dividend-price ratio by contrast is much more stable.

¹⁹ The simulated model follows Stambaugh (1999), but allows for overlapping observations as in Boudoukh et al. (2008):

$$\begin{aligned}r_{t+h} &= \alpha + \beta' x_t + u_{t+1}, \\x_{t+1} &= \theta + \phi x_t + v_{t+1},\end{aligned}$$

where h is either 1, 3, 6, or 12, and x_t is the predictor variable. Error terms are bivariate-normal. I simulate monthly returns under the null of no predictability ($\beta = 0$). Longer-horizon returns are constructed from simulated monthly returns. In 100,000 simulations, bias is measured as the mean of the predictive coefficients. Bootstrapped p -values are calculated as the percentage of simulated coefficients that are farther away from zero than the coefficients found in the data.

The approach proposed by Amihud and Hurvich (2004) confirms that the bias-adjusted estimated parameter on the corrected dividend-price ratio in the case of nonoverlapping monthly returns is almost identical to the OLS parameter, and the adjusted t -statistic is 3.40, which is marginally higher than the Hodrick (1992) t -statistic.

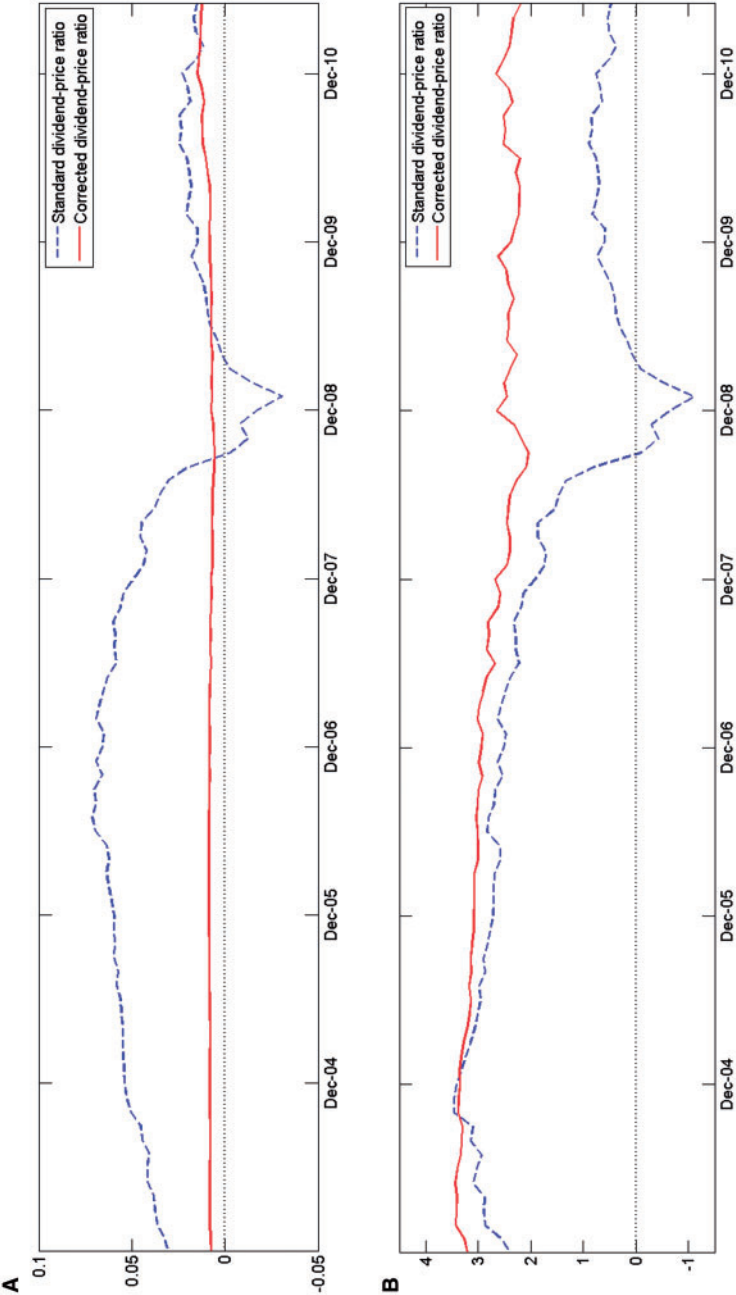


Figure 2
Parameter stability in return predictive regressions

This figure plots predictive coefficients (Panel A) and Hodrick (1992) *t*-statistics (Panel B) for univariate predictive regressions of monthly returns on either the standard dividend-price ratio or the corrected dividend-price ratio using 120-month rolling windows. The period is from January 1994 through June 2011. The first rolling window ends in December 2003.

It is always positive and statistically significant. Thus, the documented return predictability is not driven by any specific subsample.

The corrected dividend-price ratio improves the most in predicting returns during the recent financial crisis. This is intuitive, as the outbreak of the crisis completely altered expectations about future cash flows, so adjusting the standard dividend-price ratio for expected dividend growth rates should be most important during this period.

To examine return predictability in real time, I calculate the out-of-sample R^2 as in Campbell and Thompson (2008) and Goyal and Welch (2008):

$$R_{OS}^2 = 1 - \frac{MSE_{FV}}{MSE_{HA}}, \quad (22)$$

where $MSE_{FV} = \frac{1}{T-1} \sum_{t=1}^T (r_{t+1} - \hat{\mu}_t)^2$ is the mean-squared error based on the fitted values $\hat{\mu}_t$ from a predictive regression estimated through period t , and $MSE_{HA} = \frac{1}{T-1} \sum_{t=1}^T (r_{t+1} - \bar{r}_t)^2$ is the mean-squared error based on the historical average return $\bar{r}_t$ estimated through period t .²⁰ Statistical significance of results is measured by the $MSE - F$ statistic proposed by McCracken (2007). I use the period from January 1994 through December 1999 for estimation of the initial parameters and the period from January 2000 through June 2011 for calculation of the R_{OS}^2 . All out-of-sample forecasts are based on a recursive scheme using all the available information up to time t .

Because the corrected dividend-price ratio is subject to a look-ahead bias (the adjustment factor is estimated using the whole sample period), I additionally estimate a so-called no-look-ahead-bias corrected dividend-price ratio, $dp_t^{Corr-NLAB} = dp_t + \left(\frac{1}{1-\hat{\rho}_t} \right) idg_t$, where $\hat{\rho}_t$ and $\hat{\gamma}_t$ are time-varying and estimated using the same recursive scheme as in calculation of the out-of-sample R_{OS}^2 .

Table 6 reports the results. The standard dividend-price ratio with the R_{OS}^2 of 0.04% fails to predict returns in real time. The out-of-sample R_{OS}^2 for the corrected dividend-price ratio, however, is 5.94%, statistically significant at the 1% level. Thus, the corrected dividend-price ratio predicts returns both in-sample and out-of-sample. Moreover, approximately the same R_{OS}^2 is obtained with a corrected dividend-price ratio adjusted for look-ahead bias. Hence, look-ahead bias is not a concern when predicting returns with the corrected dividend-price ratio.²¹

²⁰ An in-depth discussion on the possibility of predicting returns out-of-sample is provided in a special issue of *The Review of Financial Studies* (Spiegel 2008).

²¹ The difference in the R_{OS}^2 between the two corrected dividend-price ratios is rather small because the persistence of the implied dividend growth rates $\hat{\gamma}_t$ and the linearization constant $\hat{\rho}_t$ are very stable ($\hat{\gamma}_t$ ranges from 0.8053 to 0.9059, and $\hat{\rho}_t$ is between 0.9814 and 0.9833). This makes dp_t^{Corr} and $dp_t^{NLAB-Corr}$ highly correlated at 0.9701 and almost indistinguishable.

Table 6
Out-of-sample predictability

Dependent variable: r_{t+1}^M

	dp_t	dp_t^{Corr}	$dp_t^{Corr-NLAB}$
R_{OS}^2	0.0004	0.0594***	0.0540***

This table reports out-of-sample R_{OS}^2 for predicting monthly S&P 500 returns r_{t+1}^M . Predictor variables include the standard dividend-price ratio dp_t , the monthly corrected dividend-price ratio dp_t^{Corr} , and the monthly corrected dividend-price ratio adjusted for look-ahead bias $dp_t^{Corr-NLAB}$. All variables are expressed in logs. The R_{OS}^2 is calculated over the period January 2000 through June 2011 (the period from January 1994 through December 1999 is used to make the first forecast). Statistical significance is measured according to McCracken (2007). Three asterisks (***) denote significance at the 1% level.

4.2 Alternative predictors

Following Lacerda and Santa-Clara (2010), I first consider an alternative version of the corrected dividend-price ratio, where the adjustment does not depend on the persistence of the expected dividend growth rates. I use as a proxy for the expected dividend growth either the implied dividend growth rates $dp_t^{LSC}(idg_t)$ or, as suggested by Lacerda and Santa-Clara (2010), the average of past dividend growth rates $dp_t^{LSC}(\bar{d}g_t)$. I approximate $\bar{d}g_t$ by the twelve-month trailing average of annualized monthly dividend growth rates.

Next I consider other return predictors. The cash-flow-based predictors include the earnings-price ratio ep_t and the consumption-to-wealth ratio cay_t of Lettau and Ludvigson (2001).²² Risk-related predictors include the default spread def_t , the term spread $term_t$, and the variance risk premium as implied in the S&P 500 index. The variance risk premium is defined as the difference between the variance implied in S&P 500 options and the conditional variance. Depending on the measurement of the conditional variance, I distinguish between two versions of the variance risk premium. In the first version, conditional variance is approximated by the past intradaily realized variance vrp_t^{RVH} , as in Bollerslev et al. (2009). In the second version, conditional variance is forecasted using past intradaily realized variance and the implied variance vrp_t^{FRVH} .

To control for demand imbalances in the derivatives markets, I use the skewness implied in thirty-days-to-maturity S&P 500 options $skew_t$ and the open-buy volume ob_t as in Pan and Poteshman (2006). Open-buy volume measures the difference between the total opening volume for calls and puts. Because implied interest rates could be driven by the same microstructure effects as the implied dividend growth rates, I also use the spread between six-month LIBOR and six-month implied interest rate spr_t^{LIR} .

Table 7 reports return predictability results. The particular form of the adjustment to the dividend-price ratio does not affect the inference much. At the

²² Since cay_t is available only quarterly, monthly observations are defined by the most recently available quarterly observation.

Table 7
Alternative predictors: Monthly return regressions

dp_t^{Corr}	0.0073 (3.22)			0.0083 (3.25)	0.0072 (3.10)	0.0080 (3.43)	0.0067 (2.63)
$dp_t^{LSC}(idg_t)$		0.0010 (3.09)					
$dp_t^{LSC}(\overline{dgt})$			0.0000 (0.06)				
ep_t				-0.0115 (-1.56)			
cay_t				0.1332 (1.22)			
def_t					-0.0043 (-0.38)		
$term_t$					-0.0025 (-1.05)		
vrp_t^{RVH}						0.0005 (2.29)	
vrp_t^{FRVH}						-0.0001 (-0.24)	
ob_t							0.0000 (0.95)
$skew_t$							-0.0000 (-0.00)
sp_t^{LIR}							-0.0098 (-0.63)
$adj.R^2$	0.0493	0.0491	-0.0048	0.0539	0.0490	0.1038	0.0406

This table reports regression results for predicting monthly S&P 500 returns. Predictor variables include the monthly corrected dividend-price ratio dp_t^{Corr} (defined in Table 3); the dividend-price ratio corrected according to Lacerda and Santa-Clara (2010), where expected dividend growth rates are approximated either by the implied dividend growth rates $dp_t^{LSC}(idg_t)$, or the twelve-month trailing average of annualized monthly growth rates $dp_t^{LSC}(\overline{dgt})$; the earnings-price ratio ep_t ; the consumption-to-wealth ratio cay_t (monthly observations are defined by the most recently available quarterly observation); the default spread def_t ; the term spread $term_t$; the variance risk premium defined as the difference between the variance implied by the S&P 500 options and the conditional variance, where conditional variance is approximated either by the past intradaily realized variance vrp_t^{RVH} , or the variance forecasted by past intradaily realized variance and the implied variance vrp_t^{FRVH} ; the difference between the open-buy volume for calls and puts ob_t ; the skewness implied by the S&P 500 options $skew_t$; and the spread between six-month LIBOR and the six-month implied interest rate sp_t^{LIR} . All variables are expressed in logs. Hodrick (1992) *t*-statistics are reported in parentheses. All regressions include a constant term. The period is from January 1994 through June 2011, and the frequency is monthly.

monthly horizon, both the dp_t^{Corr} and the $dp_t^{LSC}(idg_t)$ are highly statistically significant. The only difference lies in the size of the predictive coefficient.²³

The results, however, hinge on use of the implied dividend growth rates as a proxy for expected dividend growth rates. In contrast to $dp_t^{LSC}(idg_t)$, the predictive coefficient on the dividend-price ratio adjusted for the trailing average of dividend growth rates $dp_t^{LSC}(\overline{dgt})$ is insignificant, and the associated $adj.R^2$ is even lower than in the regression with the standard dividend-price ratio.²⁴

²³ At longer horizons, the dp_t^{Corr} predicts returns better than the $dp_t^{LSC}(idg_t)$.

²⁴ Untabulated results show that this result is robust to using alternative proxies for historical dividend growth rates, such as lagged monthly dividend growth rates or the moving average of ten years of monthly dividend growth rates.

The dividend-price ratio corrected for the implied dividend growth rates dp_t^{Corr} remains highly significant in a multivariate regression with other cash-flow-based predictors. It is also not affected much by inclusion of risk-related predictors, and continues to be significant when controlling for demand imbalances in the derivative markets. The associated t -statistic is never below 2.63.

Interestingly, the corrected dividend-price ratio predicts returns especially well jointly with the variance risk premium estimates. In the multivariate regression, the corrected dividend-price ratio is highly significant, with a t -statistic of 3.43. The associated $adj.R^2$ is as high as 10.38%.

Among the alternative predictors, only the variance risk premium of Bollerslev et. al (2009) vrp_t^{RVH} is significant in a multivariate regression with the corrected dividend-price ratio, with the t -statistic of 2.29. In line with Bekaert and Hoerova (2013), however, predictability results using the variance risk premium depend on how it is defined. The alternative measure of the variance risk premium vrp_t^{FRVH} turns out to be insignificant.

5. What Moves Prices?

In this section, I explore further the implications of the present value model to provide additional insights into the interpretation of results.

According to Equation (7), the dividend-price ratio varies over time because of variation in expected returns (discount rate news), variation in expected dividend growth rates (cash-flow news), or their covariation

$$\begin{aligned} var(dp_t) \simeq & \left(\frac{1}{1-\rho\delta_1} \right)^2 var(\mu_t) + \left(\frac{1}{1-\rho\gamma_1} \right)^2 var(g_t) \\ & - 2 \left(\frac{1}{1-\rho\delta_1} \right) \left(\frac{1}{1-\rho\gamma_1} \right) cov(\mu_t, g_t). \end{aligned} \quad (23)$$

Assigning half of the covariance term to expected returns and half to expected dividend growth rates yields the same results as the variance of the dividend-price ratio decomposed to two covariance terms,

$$var(dp_t) \simeq \left(\frac{1}{1-\rho\delta_1} \right) cov(dp_t, \mu_t) - \left(\frac{1}{1-\rho\gamma_1} \right) cov(dp_t, g_t). \quad (24)$$

I standardize the right-hand side of both equations by their left-hand side, so that the terms always sum up to 100%. I obtain the variance of expected returns by inverting Equation (7), and I calculate the persistence of expected returns from the OLS estimated parameter on the corrected dividend-price ratio and the linearization constant, according to Equation (9).²⁵ Using Equation (23) I also calculate the correlation between expected returns and expected dividend growth rates.

²⁵ Alternatively, the persistence of expected returns can be identified from the estimated parameter on the dividend-price ratio or the implied dividend growth rates, according to Equation (8). Both approaches yield very similar results.

Table 8
Present value parameter estimates

	(1)	(2)	(3)	(4)
$Stddev(\mu_t)$	0.0320 [0.03][0.04]	0.0505 [0.05][0.05]	0.0767 [0.07][0.09]	0.0934 [0.08][0.11]
$Stddev(g_t)$	0.1084 [0.10][0.11]	0.0643 [0.06][0.07]	0.0813 [0.08][0.08]	0.1809 [0.15][0.20]
δ_1	0.9661 [0.95][0.98]	0.9221 [0.92][0.93]	0.7487 [0.72][0.79]	0.7838 [0.76][0.84]
γ_1	0.3410 [0.29][0.40]	0.3662 [0.35][0.38]	0.3160 [0.24][0.39]	0.2155 [0.08][0.39]
Variance decomposition of the dividend-price ratio according to Eq. (23)				
Discount rate	1.0951 [1.08][1.10]	1.1901 [1.19][1.20]	1.1547 [1.12][1.18]	2.7031 [1.77][3.74]
Cash-flow	0.1297 [0.11][0.15]	0.0528 [0.05][0.05]	0.1981 [0.16][0.22]	1.0187 [0.49][1.50]
Covariance	-0.2248 [-0.24][-0.20]	-0.2429 [-0.25][-0.24]	-0.3529 [-0.39][-0.30]	-2.7218 [-4.31][-1.32]
Variance decomposition of the dividend-price ratio according to Eq. (24)				
Discount rate	0.9827 [0.98][0.98]	1.0687 [1.07][1.07]	0.9783 [0.97][0.99]	1.3422 [1.11][1.52]
Cash-flow	0.0173 [0.02][0.02]	-0.0687 [-0.07][-0.07]	0.0217 [0.01][0.03]	-0.3422 [-0.52][-0.11]
Correlation between expected returns and expected dividend growth				
$\rho_{\mu_t g_t}$	0.2975 [0.29][0.31]	0.4845 [0.48][0.49]	0.3681 [0.34][0.39]	0.7361 [0.67][0.89]

This table reports the present value parameter estimates. Delta one δ_1 is annual persistence of expected returns μ_t , and gamma one γ_1 is annual persistence of expected dividend growth rates g_t . Expected dividend growth rates are approximated by lagged annual dividend growth rates in Columns (1) through (3) and by implied dividend growth rates in Column (4). The means for the parameters are estimated across 12 successive nonoverlapping annual samples. The corresponding 25th and 75th percentile are reported in brackets. Variance decomposition of the dividend-price ratio is based on either Equation (23) or Equation (24). The period is from January 1926 through June 2011 in Column (1), from January 1946 through June 2011 in Column (2), and from January 1994 through June 2011 in Columns (3) and (4).

To avoid problems with overlapping observations, I follow the typical approach in the literature and rely on nonoverlapping annual observations. Because the time period in the main analysis is rather short, but the data are available monthly, I estimate all parameters 12 times using successive annual samples starting at the end of each month. I report the mean along with the 25th and the 75th percentiles for the estimated parameters.

For a comparison with the implied dividend growth rates, I first take lagged annual dividend growth rates as a proxy for expected dividend growth rates. This also lets me examine how the sample period analyzed in this paper compares to longer time periods.

Table 8 reports the results. Using lagged dividend growth rates as a proxy for expected dividend growth rates, I find that almost all variation in the dividend-price ratio is driven by the discount rate news, consistent with the findings of, for example, Cochrane (2008a), Binsbergen and Koijen (2010), Koijen and Van Nieuwerburgh (2011), and Cochrane (2011). I also find that the results are very similar across periods starting in 1926, 1946, and 1994. Thus, the main period

analyzed starting in 1994, although rather short, is by and large comparable to longer periods.

In the main period, expected returns tend to be more persistent than expected dividend growth rates, 0.75 and 0.32 on average, respectively. Both are positively correlated at 0.37 on average, in line with Menzly et al. (2004) and Lettau and Ludvigson (2005). According to Equation (23), variation in expected returns and expected dividend growth rates account on average for 115% and 20%, respectively, of the variation in the dividend-price ratio, while the covariance term accounts for -35% of such variation. The alternative decomposition in Equation (24) assigns 98% of dividend-price ratio variance to covariance with expected returns, and 2% to covariance with expected dividend growth rates. The estimated parameters across alternative annual samples are fairly stable as reported by the rather small interquartile ranges.

When I replace lagged dividend growth rates with implied dividend growth rates as a measure of expected dividend growth rates, the mean persistence of expected returns rises slightly to 0.78, and the mean persistence of expected dividend growth rates declines to 0.22. Most notably, the mean correlation between expected returns and expected dividend growth rates increases from 0.37 to as much as 0.74. At the same time, differences between estimated parameters across different annual samples become much more pronounced. The increase in the mean correlation, however, cannot be explained simply by a wider interquartile range. The 25th and the 75th percentiles are 0.67 and 0.89, respectively, compared to 0.34 and 0.39, when lagged dividend growth rates are used as a proxy for expected dividend growth rates.²⁶

This increase in the correlation implies that the covariance term in Equation (23) has a much more important effect on reducing the variance of the dividend-price ratio. Indeed, the covariance term accounts on average for as much as -272% of the variation in the dividend-price ratio (the 25th and the 75th percentiles are -431% and -132%). The expected returns account on average for 270% of the dividend-price ratio variance (at the 25th and the 75th percentiles, 177% and 374%). Finally, 102% of the dividend-price ratio variance is on average due to time-varying expected dividend growth rates (at the 25th and the 75th percentiles, 49% and 150%). Thus, expected returns and expected dividend growth rates vary much more than is apparent from variation in the standard dividend-price ratio.

²⁶ Note that the increase in the correlation is not due simply to greater variability of implied dividend growth rates. Two sets of results confirm this. First, I use a Hodrick-Prescott filter to filter out the trend component of the implied dividend growth rates, which I then use as a proxy for expected dividend growth rates. In untabulated results I find that, although the trend component is substantially less volatile than implied dividend growth rates, the estimated correlation between expected returns and expected dividend growth rates in this case is even higher, 0.86 on average. In the second exercise, I contaminate implied dividend growth rates with a layer of normally distributed noise with volatility equal to 50% of the standard deviation of implied dividend growth rates. In 100,000 simulations, I find that the addition of noise barely changes the correlation estimates. The average correlation stays largely unchanged at 0.74.

When the covariance term is split between expected returns and expected dividend growth rates, covariance with expected returns prevails and accounts on average for 134% of the variation in the dividend-price ratio (at the 25th and the 75th percentiles, 111% and 152%). The rest, -34% of the dividend-price ratio variation, is accounted for by covariance with the dividend growth rates (at the 25th and the 75th percentiles, -52% and -0.11%).^{27, 28}

All in all, although the time period is relatively short and there is substantial variation between the annual samples, expected dividend growth rates appear much more correlated with expected returns, in comparison with the benchmark case and the typical values reported in the literature. This suggests that the offsetting effects of expected returns and expected dividend growth rates on the dividend-price ratio documented by Menzly et al. (2004) and Lettau and Ludvigson (2005) are magnified, and there is a more severe omitted variable bias in the standard dividend-price ratio predictive regression than implied in recent studies.

6. Announced Dividends

As companies typically announce dividends one to three months in advance, we can use this information to predict aggregate dividend growth rates. Indeed, this is probably the simplest way to forecast dividends. Therefore, it is instructive to compare dividend growth rates based on announced dividends with the implied dividend growth rates.

There are good reasons to believe that dividends implied in derivative markets encompass a richer set of information than announced dividends. Several studies document that derivatives subsume information beyond publicly available information.²⁹ While announced dividends are observed, however, implied dividends are estimated and thus subject to measurement errors.

Further, because information on announced dividends is public, dividend growth predictability using announced dividends is rather mechanical and short-horizon oriented. Thus, we may expect that the part of dividend growth rates that cannot be predicted using information on announced dividends is most relevant for price movements. Finally, as already discussed, the fact that part of dividends can be predicted using publicly available information on announced

²⁷ According to the present value model, the negative covariance between the dividend-price ratio and the expected dividend growth rates emerges because expected returns and expected dividend growth rates are highly correlated.

²⁸ The results are not very sensitive to different levels of the persistence of expected dividend growth rates. Rather than estimate the persistence of implied dividend growth rates, I set it at anywhere between 0.2 and 0.5 (typical values reported in the literature). In untabulated results, I find that discount rate news in this case accounts for anywhere between 127% and 143% of the dividend-price ratio variance, according to Equation (24). Cash-flow news accounts for anywhere between -27% and -43% of such variation. The correlation between expected returns and expected dividend growth rates is between 0.74 and 0.84.

²⁹ For example, SPX implied volatility is a more efficient predictor of future realized volatility than past realized volatility (Jiang and Tian 2005).

dividends can explain the lack of a dividend risk premium in short-maturity dividend claims.

To analyze these issues, I use twelve-month indicated dividends, calculated by annualizing the announced dividends of all stocks in the S&P 500. If no dividend is announced, the most recent dividend payment is projected over the next year. I define indicated dividend growth rates dg_t^{IND} as the log difference between twelve-month indicated dividends and the twelve-month trailing sum of realized dividends.

I consider predicting first-year dividend growth rates $\Delta d_{t,t+12}^A = \log[D_{t+12}^{12}/D_t^{12}]$ and second-year dividend growth rates $\Delta d_{t+12,t+24}^A = \log[D_{t+24}^{12}/D_{t+12}^{12}]$. I also explore whether residual dividend growth rates, defined as the difference between first-year dividend growth rates and the indicated dividend growth rates $\Delta d_{t,t+12}^A - dg_t^{IND}$, help to explain returns.

The results are reported in Table 9. For data availability reasons, the sample period runs from February 1995 through June 2011. The indicated dividend growth rates predict first-year dividend growth rates with an $adj.R^2$ as high as 72.52% and a t -statistic of 4.63. A substantial part of dividends is indeed easy to predict, and hence riskless. This explains why the dividend risk premium is absent in short-maturity dividend claims.³⁰

Such predictability is short-lived, however, and has no implications for pricing. That is, indicated dividend growth rates explain merely 0.22% of the variation in the second-year dividend growth rates, with an insignificant t -statistic of 1.12. Indicated dividend growth rates also do not help in predicting monthly returns. The $adj.R^2$ actually drops when indicated dividend growth rates are added to the standard dividend-price ratio return predictive regression.³¹

Adding residual dividend growth rates to the standard dividend-price ratio monthly return regression, by contrast, increases the $adj.R^2$ from 0.62% to as much as 11.90%. Although residual dividend growth rates are not observed in real time and thus cannot be used for forecasting, this result confirms that the part of dividend growth rates that cannot be predicted by publicly available information on indicated dividends contaminates the dividend-price ratio the most.

In comparison with indicated dividend growth rates, implied dividend growth rates explain a smaller part of the variation in the first-year dividend growth rates, with an $adj.R^2$ of 19.59%. Implied dividend growth rates, however, are statistically significant predictors for first- and second-year dividend growth rates, with t -statistics of 4.94 and 2.14. Further, implied dividend growth rates remain statistically significant predictors for first- and second-year dividend

³⁰ In untabulated results, I show that indicated dividend growth rates predict semiannual dividend growth rates in the same period with an $adj.R^2$ of 90.69%.

³¹ In untabulated results, I show that the indicated dividend growth rates also fail to predict longer-horizon returns.

Table 9
Announced dividends
Panel A: Predicting dividend growth

First-year growth rate			Second-year growth rate		
		$\Delta d_{t,t+12}^A$		$\Delta d_{t+12,t+24}^A$	
dp_t		-0.0291 (-0.55)			-0.0014 (-0.04)
idg_t	0.2019 (4.94)	0.0808 (2.48)	0.0705 (2.14)		0.0655 (2.53)
dg_t^{IND}		1.3635 (4.63)		0.1556 (1.12)	0.0546 (0.31)
$adj.R^2$	0.1959	0.7252	0.7441	0.0187	0.0022 0.0081
Panel B: Predicting monthly returns r_{t+1}^M					
dp_t		0.0204 (1.15)	0.0084 (0.42)	0.0211 (1.23)	0.0290 (1.74)
idg_t			0.0560 (2.50)		0.0227 (1.03)
dg_t^{IND}			0.0100 (0.14)		
$\Delta d_{t,t+12}^A - dg_t^{IND}$				0.3349 (2.81)	0.2991 (2.40)
$adj.R^2$		0.0062	0.0476	0.0009	0.1190 0.1204

Panel A reports regression results for predicting first-year dividend growth rates $\Delta d_{t,t+12}^A = \log[D_{t+12}^{12}/D_t^{12}]$ and second-year dividend growth rates $\Delta d_{t+12,t+24}^A = \log[D_{t+24}^{12}/D_{t+12}^{12}]$, where D_t^{12} is the twelve-month trailing sum of dividends. Panel B reports regression results for predicting monthly S&P 500 returns. Predictor variables include the standard dividend-price ratio dp_t , the implied dividend growth rates idg_t , and the indicated dividend growth rates dg_t^{IND} . All variables are expressed in logs. Hodrick (1992) t -statistics accounting for the overlap are reported in parentheses. The period is from February 1995 through June 2011, and the frequency is monthly.

growth rates in multivariate regressions with the standard dividend-price ratio and the indicated dividend growth rates. Thus, they reveal part of the variation in future dividend growth rates that cannot be predicted by indicated dividends. As these rates contaminate the dividend-price ratio the most, this result explains why implied dividend growth rates predict returns better than indicated dividend growth rates even though indicated dividend growth rates explain a greater part of the variation in first-year dividend growth rates. I find accordingly that implied dividend growth rates become insignificant return predictors when controlling for residual dividend growth rates.

7. Conclusions

It has long been accepted that changes in expected dividend growth rates make the standard dividend-price ratio less able to predict returns. I address this issue using a novel estimate for expected dividend growth rates extracted from derivative markets (index options and futures). I show that dividend growth rates implied by derivative markets reliably forecast future dividend growth rates, and that the dividend-price ratio corrected for variation in implied dividend growth rates predicts S&P 500 returns substantially better than the standard dividend-price ratio. The forecasting relation between the corrected

dividend-price ratio and returns is statistically significant both in-sample and out-of-sample.

Investigating the implications of present value relations, I find that time-varying expected dividend growth rates are important for dividend-price movements especially because they are highly correlated with expected returns. This positive covariation makes the standard dividend-price ratio very smooth, weakening its ability to predict returns and dividend growth rates. Once the standard dividend-price ratio is corrected for variation in expected dividend growth rates, it becomes more volatile and less persistent, and better able to predict returns.

The results are robust to a number of variations, and are not driven by seasonality of dividend payments or dividend risk premium.

Appendix A. Derivation of the present value model

Define log return r_{t+1} , log dividend growth rate Δd_{t+1} , and log dividend-price ratio dp_t as

$$r_{t+1} = \log \left[\frac{P_{t+1} + D_{t+1}}{P_t} \right], \quad \Delta d_{t+1} = \log \left[\frac{D_{t+1}}{D_t} \right], \quad dp_t = \log \left[\frac{D_t}{P_t} \right]. \quad (A1)$$

Let $\mu_t = E_t(r_{t+1})$ be conditional expected return and let $g_t = E_t(\Delta d_{t+1})$ be conditional expected dividend growth rate. Assume that μ_t and g_t follow AR(1) processes

$$\mu_{t+1} = \delta_0 + \delta_1(\mu_t) + \varepsilon_{t+1}^\mu, \quad (A2)$$

$$g_{t+1} = \gamma_0 + \gamma_1(g_t) + \varepsilon_{t+1}^g, \quad (A3)$$

$$\Delta d_{t+1} = g_t + \varepsilon_{t+1}^d, \quad (A4)$$

where ε_{t+1}^μ , ε_{t+1}^g and ε_{t+1}^d are zero mean errors.

Rewrite log returns as

$$r_{t+1} = dp_t + \Delta d_{t+1} + \log[1 + \exp(-dp_{t+1})]. \quad (A5)$$

Use a first-order Taylor expansion to linearize $\log[1 + \exp(-dp_{t+1})]$ around $\bar{dp} = E(dp_t)$:

$$\log[1 + \exp(-dp_{t+1})] \simeq \log[1 + \exp(-\bar{dp})] + \frac{\exp(-\bar{dp})}{1 + \exp(-\bar{dp})} [-dp_{t+1} + \bar{dp}]. \quad (A6)$$

Define $\rho = \frac{\exp(-\bar{dp})}{1 + \exp(-\bar{dp})}$ and $\kappa = \log[1 + \exp(-\bar{dp})] + \rho\bar{dp}$, such that

$$\log[1 + \exp(-dp_{t+1})] \simeq \kappa - \rho dp_{t+1}. \quad (A7)$$

Plug (A7) into (A5) to get an expression for the one-period return:

$$r_{t+1} \simeq \kappa + dp_t + \Delta d_{t+1} - \rho dp_{t+1}. \quad (A8)$$

Iterate equation (A8) forward:

$$\begin{aligned} dp_t &\simeq -\kappa + \rho dp_{t+1} + r_{t+1} - \Delta d_{t+1} \\ &\simeq -\kappa + \rho(-\kappa + \rho dp_{t+2} + r_{t+2} - \Delta d_{t+2}) + r_{t+1} - \Delta d_{t+1} \\ &\simeq -\kappa - \kappa\rho + \rho^2 dp_{t+2} + r_{t+1} + \rho r_{t+2} - \Delta d_{t+1} - \rho \Delta d_{t+2} \\ &\simeq -\frac{\kappa}{1-\rho} + \rho^\infty dp_{t+\infty} + \sum_{j=0}^{\infty} \rho^j (r_{t+1+j} - \Delta d_{t+1+j}). \end{aligned} \quad (A9)$$

Assume that $\lim_{j \rightarrow \infty} \rho^j dp_{t+j} = 0$ to obtain the Campbell and Shiller (1988) approximation for the log dividend-price ratio (since the relationship holds ex-ante and ex-post, an expectation operator can be added to the right-hand side).

$$dp_t \simeq -\frac{\kappa}{1-\rho} + E_t \sum_{j=0}^{\infty} \rho^j (r_{t+1+j}) - E_t \sum_{j=0}^{\infty} \rho^j (\Delta d_{t+1+j}). \quad (\text{A10})$$

Iterate equations (A2) and (A3) forward:

$$E_t(r_{t+1+j}) = \delta_0 \frac{1-\delta_1^j}{1-\delta_1} + \delta_1^j \mu_t, \quad (\text{A11})$$

$$E_t(\Delta d_{t+1+j}) = \gamma_0 \frac{1-\gamma_1^j}{1-\gamma_1} + \gamma_1^j g_t. \quad (\text{A12})$$

Work out the expectations of $E_t \sum_{j=0}^{\infty} \rho^j (r_{t+1+j})$ and $E_t \sum_{j=0}^{\infty} \rho^j (\Delta d_{t+1+j})$.

$$\begin{aligned} E_t \sum_{j=0}^{\infty} \rho^j (r_{t+1+j}) &= \sum_{j=0}^{\infty} \rho^j \left(\delta_0 \frac{1-\delta_1^j}{1-\delta_1} + \delta_1^j \mu_t \right) \\ &= \frac{\delta_0}{1-\delta_1} \sum_{j=0}^{\infty} \rho^j - \frac{\delta_0}{1-\delta_1} \sum_{j=0}^{\infty} \rho^j \delta_1^j + \mu_t \sum_{j=0}^{\infty} \rho^j \delta_1^j \\ &= \frac{\delta_0}{(1-\delta_1)(1-\rho)} - \frac{\delta_0}{(1-\delta_1)(1-\rho\delta_1)} + \mu_t \left(\frac{1}{1-\rho\delta_1} \right), \end{aligned} \quad (\text{A13})$$

$$\begin{aligned} E_t \sum_{j=0}^{\infty} \rho^j (\Delta d_{t+1+j}) &= \sum_{j=0}^{\infty} \rho^j \left(\gamma_0 \frac{1-\gamma_1^j}{1-\gamma_1} + \gamma_1^j g_t \right) \\ &= \frac{\gamma_0}{1-\gamma_1} \sum_{j=0}^{\infty} \rho^j - \frac{\gamma_0}{1-\gamma_1} \sum_{j=0}^{\infty} \rho^j \gamma_1^j + g_t \sum_{j=0}^{\infty} \rho^j \gamma_1^j \\ &= \frac{\gamma_0}{(1-\gamma_1)(1-\rho)} - \frac{\gamma_0}{(1-\gamma_1)(1-\rho\gamma_1)} + g_t \left(\frac{1}{1-\rho\gamma_1} \right). \end{aligned} \quad (\text{A14})$$

Finally, insert Equations (A13) and (A14) in the Campbell and Shiller (1988) present value identity to find the dividend-price ratio

$$dp_t \simeq \varphi + \mu_t \left(\frac{1}{1-\rho\delta_1} \right) - g_t \left(\frac{1}{1-\rho\gamma_1} \right), \quad (\text{A15})$$

where $\varphi = -\frac{\kappa}{1-\rho} + \frac{\delta_0}{(1-\delta_1)(1-\rho)} - \frac{\delta_0}{(1-\delta_1)(1-\rho\delta_1)} - \frac{\gamma_0}{(1-\gamma_1)(1-\rho)} + \frac{\gamma_0}{(1-\gamma_1)(1-\rho\gamma_1)}$.

Appendix B. Bias in the persistence estimates

To analyze the impact of measurement errors (noise) and overlapping observations on the persistence of the implied dividend growth rates, I start by simulating expected dividend growth rates and realized dividend growth rates according to Equations (5) and (6). I assume a monthly $AR(1)$ coefficient of 0.9 and normally distributed innovations. I set volatilities of innovations to expected dividend growth rates and realized dividend growth rates to 0.004 and 0.0004, respectively. I match the length of the simulated series with the length of the sample period in the main analysis ($N=210$). I then calculate the twelve-month trailing sum of dividends D_t^{12} and

Table B.1
Bias in the persistence estimates

Simulated process with $AR(1)$ of 0.9	No noise	With noise $[0.2 * std(E(D_t^6))]$	With noise $[0.3 * std(E(D_t^6))]$
<i>Method A:</i>			
Monthly $AR(1)$	0.9704	0.6616	0.4968
<i>Method B:</i>			
12th root of the annual $AR(1)$	0.9092	0.8713	0.8408

This table reports simulation results for analysis of the measurement errors (noise) and overlapping observations on the persistence of implied dividend growth rates. Simulations of implied dividend growth rates are based on the assumptions of the present value model and empirical procedures for estimating implied dividend growth rates. Persistence is measured either as the simple $AR(1)$ of the monthly simulated series (Method A), or as the 12th root of the annual $AR(1)$ averaged across 12 successive annual samples (Method B). Column (1) reports results for the base case. In Columns (2) and (3), a layer of measurement errors (noise) is added to the simulated expected dividends.

expected dividends over the next 6 months $E(D_t^6)$. Simulated implied dividend growth rates are defined as $\log(2 \times E(D_t^6)/D_t^{12})$.

I measure the persistence of the simulated monthly implied dividend growth rates either as a simple $AR(1)$ (Method A), or as as the 12th root of the annual $AR(1)$ averaged across 12 successive annual samples (Method B).

Table B.1 reports results for 100,000 simulations. Column (1) reports results for the base case, subject to the overlapping observations, but without noise. In Columns (2) and (3), I add noise to the $E(D_t^6)$ in the form of a normally distributed variable with mean zero and standard deviation equal to either 20% or 30% of the standard deviation of the simulated $E(D_t^6)$.

Without noise, the estimated monthly $AR(1)$ of the simulated series is upward biased due to the overlapping observations at 0.97. With noise, however, the effect of the noise prevails, and the persistence estimates are consequently downward biased. When noise equals $0.3 \times std(E(D_t^6))$, Method A underestimates monthly $AR(1)$ by as much as 44.80% (monthly persistence is 0.50). This bias, however, is largely diminished by inferring monthly persistence from the annual values. Method B underestimates monthly $AR(1)$ by merely 6.58% (monthly persistence of 0.84).

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